Park Smart Pro - Smart Parking Solution for Telangana

A Course Project Report

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Title

Park Smart Pro – Smart Parking Solution for Telangana

ABSTRACT:

Park Smart Pro is a digital application designed as a smart parking management system in the city of Telangana to help to alleviate the problems associated with obtaining parking spaces in urban locations with increasing numbers of vehicles and limited parking availability. Due to the increase in both vehicle numbers as well as limited parking, finding locations to park your vehicle have become difficult and time-consuming. This application uses real-time data to allow users the ability to see available spaces, instantly book a space, print a digital ticket, and pay for parking at the time of exiting based on the amount of time you parked.

In addition to a customer user role, Park Smart Pro also has an admin user role to allow the admin to view bookings made by customers, manage parking locations, track parking revenue generated, and see the availability of parking locations using dashboards. The application is created using HTML, CSS, JavaScript, and Local Storage technology and provides a simple, responsive solution through the use of a web browser without supporting any additional frameworks. By providing customers the tools to find parking easier Park Smart Pro will assist in reduction of traffic congestion, time wasted and fuel consumed in searching for parking, and improving the efficiency of the overall parking experience. By replacing manual systems with a digital approach, Park Smart Pro is aiding in the development of smart cities.

Keywords: Smart Parking System, Parking Management, Real-Time Slot Availability, Digital Ticket Generation, Online Booking, Exit Payment System, Admin Dashboard, Customer Module, Web Application, HTML, CSS, JavaScript, Local storage, Smart City Solution, Telangana Parking System, Vehicle Slot Booking

OBJECTIVES:

Park Smart Pro intends to develop a functional, efficient and intuitive Parking Management Application. Key features of the application include: checking real-time availability prior to arriving at a destination; reducing idle time; and eliminating unnecessary vehicle congestion.

The application also provides users with the ability to easily and instantaneously book a parking space and receive a digital parking ticket (via email), as well as providing users with an easy-to-follow exit payment process (i.e., only charging the user for the length of time that the user has actually parked).

Park Smart Pro's objective is to offer multiple parking locations across the state of Telangana and to offer separate administrative dashboards for admins to monitor bookings, usage of their parking slots, and revenue generated from the parking slots being used. The Park Smart Pro system reduces the number of mistakes that are currently made within the parking management industry and improves the overall efficiency of parking.

Primary Objectives

Park Smart Pro focuses on creating a smart solution for parking that enables the users to find and book parking spaces easily. Park Smart Pro will give you live slot availability, allowing you to know which parking spots are available even before reaching there.

Book your parking spot in no time, and you will get an electronic ticket instantly. Future plans include making an automated payment process where you get charged based on the time you parked there.

Secondary Objectives

The system's objectives include aiding in reducing vehicle accident rates and fuel consumption associated with manually searching for available or occupied parking spaces, improving customer convenience through digital mediums associated with the overall parking experience, minimizing manual error through having an accurate electronic record of every parking transaction (including the booking process); improving the administration of parking through dashboards and reporting, and aiding the development of "smart" cities by providing a modern solution to the problem of parking.

INTRODUCTION:

In urban areas, parking management can be difficult due to the increase of vehicles and limited amounts of available parking. Traditional parking systems are primarily manual, slow, and inefficient. Drivers commonly spend long periods looking for places to park, contributing to traffic congestion, wasting fuel, and becoming frustrated.

Park Smart Pro is a smart parking solution developed specifically for Telangana. With the Park Smart Pro app, users can view availability of each of the parking locations prior to arriving; they can reserve spaces instantly, allow them to check-out digitally (with a digital parking ticket), and provide payment for parking once they leave using their actual time parked.

The system also provides enhancements for administration through reporting on bookings, availability of parking spaces, and revenue generated. Areas currently covered by the system include, but aren't limited to, Hyderabad, Warangal, Karimnagar, and Nizamabad. This project will improve parking management efficiency, save time, reduce tasks, and further support the smart city infrastructure.

Background and Motivation

People are increasingly frustrated by trying to find a place to park. Given the rapid growth of cities and increase in the number of vehicles on the roads, parking management presents a challenge for many urban areas. Traditional methods of managing parking spaces usually are manual in nature; therefore, a user would physically search for available parking space on their own. This process takes time, causes traffic congestion, increases the amount of fuel consumed and results in driver frustration.

Poor management of available parking spaces significantly affects how people use and travel through commercial areas such as shopping malls, train stations and other commercial developments. With cities evolving into digital environments and developing smart cities, the demand for smart parking systems is growing.

If people can expect to receive quicker, more convenient services through any aspect of city living, including how to find a parking space which can reduce stress from not knowing where to park before arriving, the demand for this type of system will also continue to grow. Real-time parking availability, instant reservation capabilities, e-tickets and a streamlined payment process will greatly enhance the parking experience for individuals.

The Park Smart Pro project intends to provide a user-friendly, efficient way to manage car parking in the state of Telangana through automated processes that will reduce the amount of manual labour involved, increase transparency of all operations, and allow for greater management flexibility for both end users as well as administration staff. Park Smart Pro supports the continued evolution of smart urban transportation and improved parking management through the use of Web-based technology and electronic processing.

PROBLEM STATEMENT:

Traditional parking systems are inefficient and bothersome to users. They generate backups of cars waiting for open spaces and waste time and fuel due to the excessive amount of driving around the block looking for open parking.

Users looking for parking do not have a means to reserve a spot prior to their arrival or generate a digital parking ticket. Parking attendants have trouble managing customer parking space, resulting in unhappy customers.

As urban areas continue to grow, so do the difficulties associated with finding parking. Cities are in desperate need of a smart parking solution that would provide real time availability of open spaces, allow for immediate booking, provide simple payment processing and enable more optimal management of parking assets.

Park Smart Pro is providing a way to improve the parking experience using technology based solutions for many of the identified challenges.

Challenges

Finding available parking spaces in real-time is a major issue for parking management. Therefore, searching for open parking lots is an inefficient use of time and contributes to increased traffic congestion and fuel usage. This issue can increase in severity due to overly congested and busy downtown areas and can negatively affect daily transit to and from those areas.

Booking and payment management via manual methods is another challenge for parking management, as traditional methods do not provide the necessary digital records for ensuring accuracy, clarity, and providing an overall good customer experience. Cancellation of reservations and calculating the monetary amounts for completed reservations become extremely difficult due to the lack of a well-organized system.

Administering bookings, slot usage, and revenue via a central system is yet another problem associated with parking management as there are no centralized systems for monitoring the bookings and usage of each individual parking lot, this can make it very easy for parking attendants to mismanage available slots or multiple locations.

Another important part of the project from a technical standpoint is maintaining accurate reservation records and providing duplicate avoidance features. Local Storage will be utilized for this project, which will limit the method of storing data in the browser and will limit usage to single-user mode, thus limiting the ability to ensure data integrity and adequate security at this time. Finally, the lack of support for real-time, multi-user action will be a development task for future implementation of this project.

EXISTING SYSTEM

Most current parking systems are operated manually. Users must go to parking lots in person to see if a parking space is open. This takes time and gas as users cannot see the parking space from

anywhere else. There's no way for users to book their reservation digitally, and there's no digital, real-time way to view the parking lot.

Through the manual payment method for parking, there is a large amount of confusion around record keeping. No electronic ticketing system exists, and users find it difficult to cancel bookings. When it comes to revenue and parking space monitoring, admins may struggle as well because of the absence of centralized dashboard for managing those functions.

Therefore, the current parking system is slow, inefficient, and creates inconvenience to both users and administrators.

PROPOSED SYSTEM

A new smart parking management system is currently in development and will be known as Park smart Pro. The system allows customers to view real-time availability of parking spaces and make instant reservations through the Internet, while also receive an electronic parking pass.

Customers will have the ability to pay for their stay when exiting based on how long they parked in a certain area. Customers will also be able to cancel their reservation and receive a refund.The Park smart Pro system will support multiple cities and multiple parking facilities across Telangana.

The Park smart Pro system can provide administrators with a dashboard to manage reservation and usage data for each parking facility, as well as search for vehicle information, and gain insights about the revenue generated for each location. This will promote increased transparency, reduce time spent manually managing and improve efficiency in the management of the entire parking system.

TECHNOLOGIES USED

HTML5 – Used to create the basic structure and layout of all web pages in the project.

CSS3 – Used for styling, responsive design, animations, and improving the visual appearance of the application.

JavaScript – Used to implement logic, interactivity, booking operations, payment handling, and dashboard functionality.

Local Storage API – Used to store booking details and maintain data persistence in the browser.

Visual Studio Code – Used as the main code editor for writing and managing the project files.

Google Chrome – Used for testing, debugging, and running the web application.

Google Fonts (Poppins) – Used for modern and clean typography in the user interface.

CUSTOMER MODULE

With the use of the module for Customer, it is possible to reserve a space for parking with great ease. Making reservations can be accomplished by customers via the website as a variety of cities, sites, and space availability can be found at their fingertips while providing a vehicle's details. In addition, any confirmed reservation will automatically generate a digital ticket containing all relevant details and a unique booking identification number. Also, customers can complete their payment while exiting or cancel if needed after they book.

ADMIN MODULE

Managing a complete parking solution is made possible with the android admin module. There are 2 types of admin roles that you can have in the admin module, a Main Administrator (which manages all the locations, bookings and revenue), and a Location Administrator (which manages the specific location assigned to that location administrator). Admins can view the dashboards, check to see slots of parking are available, and search booking records on their respective devices.

PAYMENT MODULE

When leaving the parking lot, the payment module is used to calculate the charges based on the hourly rate of parking by entering both a customer's booking ID and how long they parked. A payment must be processed for the system to mark the booking as paid and record the appropriate revenue.

TICKET GENERATION MODULE

Successful use of the Ticket Generation Module will produce a parking ticket in digital form. The ticket will contain: booking ID, vehicle number, where the car is parked, date/time parked, charge for parking space. This ticket will assist customers in making payments and will allow for tracking of proper records by administrators.

METHODOLOGY:

Using a straightforward and logical approach to developing Park Smart Pro, we were able to implement the smart parking system without problems. We started by reviewing the parking issues faced in cities and determining what is required for any smart parking system (e.g., slot booking, digital ticketing, an electronic payment system, and admin monitoring). Using this information, the system design and workflow were created.

For the front-end, we created a user-friendly and responsive website using HTML and CSS. To complete the user interface functionality (i.e., booking logic, payment calculations, issuing tickets, and updating the admin dashboard), we used JavaScript. We chose to use Local Storage to store

booking data and maintain historical data without the need for a backend database. The final phase was to conduct usability testing to ensure that the system functioned properly for the end-users.

IMPLEMENTATION

The first stage of developing Park Smart Pro is the landing page containing the hero section, selection of the city, grouping of feature blocks, and login options for both end-users and Admins.

After developing the landing page, the customer page was developed to allow users to select their city, from which they can choose the parking garage they want to reserve a slot within (by searching for parking), enter their vehicle details, and check slot availability.

Once the customer has made their reservation using the booking module (after making a successful reservation), a single reservation ID and ticket will have been created. The payment module will have determined the parking fee by the number of hours parked for that vehicle. The cancellation module will allow the user to cancel their active reservation.

The admin module consists of two levels of administrator login authentication (Main Administrator and Location Administrator/Storage Administrator). The administrator dashboards are used to present information such as bookings, parking garage availability (space) and revenue accounting. All data is stored within Local Storage to allow quick access to that information for reference.

TRAINING AND TESTING

Testing was performed on all of the main parts of the program such as customer booking, payment processing, ticket creation, cancelling bookings after they were made, and managing the Admin Dashboard. Testing included multiple types of tests to assess both valid and invalid user input (e.g. multi-booking, accurate payment, etc.), availability of booked slots, correct updates of payment calculations, and checking for duplicate bookings. Also, testing was conducted using Google Chrome for responsive design and performance during execution. The application was visually checked to verify that no errors occurred and that validation occurred correctly at each step of the booking/payment process (i.e. correct updating of dashboard data follows successful booking or payment).

PERFORMANCE EVALUATION

Park Smart Pro's performance was evaluated for speed, usability, and accuracy of all the booking functions performed on this system. It provides confirmation of all bookings performed immediately with the generation of digital tickets also being done immediately.

Local Storage is used in this system, which enables fast retrieval and updates of data within your web browser.

The payment module accurately calculates the costs of parking based upon the type of vehicle parked and the duration of parking. Each time a user makes or cancels a booking, the admin dashboard is updated immediately, allowing for real-time visibility of the records.

The Park Smart Pro application operates well in all modern browsers, offering customers a positive experience with intuitive usability and responsive design to enhance customer service levels and provide administrators with better control.

SYSTEM ARCHITECTURE

Park smart Pro is designed to have two main types of users: the Customer and the Administrator. They interact with a web application that has been built on the front-end with HTML, CSS and JavaScript.

The current version of the system does not contain a back-end server because all data is stored in Local Storage.

The customer's side of the application allows a customer to select a city, book a parking spot, receive a digital ticket, pay and cancel their reservation. The administrator's side allows him or her to track all bookings, the usage of each parking spot, and the system's revenues by way of various dashboards.

All of the booking records are stored as objects in Local Storage, which are then dynamically updated using JavaScript. This allows for a simple, lightweight and flexible architecture for future enhancements to the back-end using MongoDB and hosted on the cloud.

OBSERVATION

The development and testing of Park Smart Pro demonstrated that the process of digital booking reduced the amount of time required to search for a parking space. The customers booked their slots quickly and received a clear confirmation of their booking via digital tickets.

With Administration Dashboard, real-time booking statistics and usage were easily monitored, and the Revenue Tracking System was much easier to track than using a manual system. Local Storage was used for fast and efficient frontend implementation; however, Local Storage does have limitations in regard to multi-user usage.

Additionally, responsive UI design was shown to increase usability, making both the customer experience and the administration experience easier to use.

PERFORMANCE ANALYSIS

Park Smart Pro provides a more efficient parking method than older individualized systems. The utilization of real-time parking slot status, eliminated excess traffic of drivers looking for parking spots, and reduced carbon emissions through less fuel usage.

With instant reservations and digitally generated tickets, customers waiting for a space will have shorter wait times.

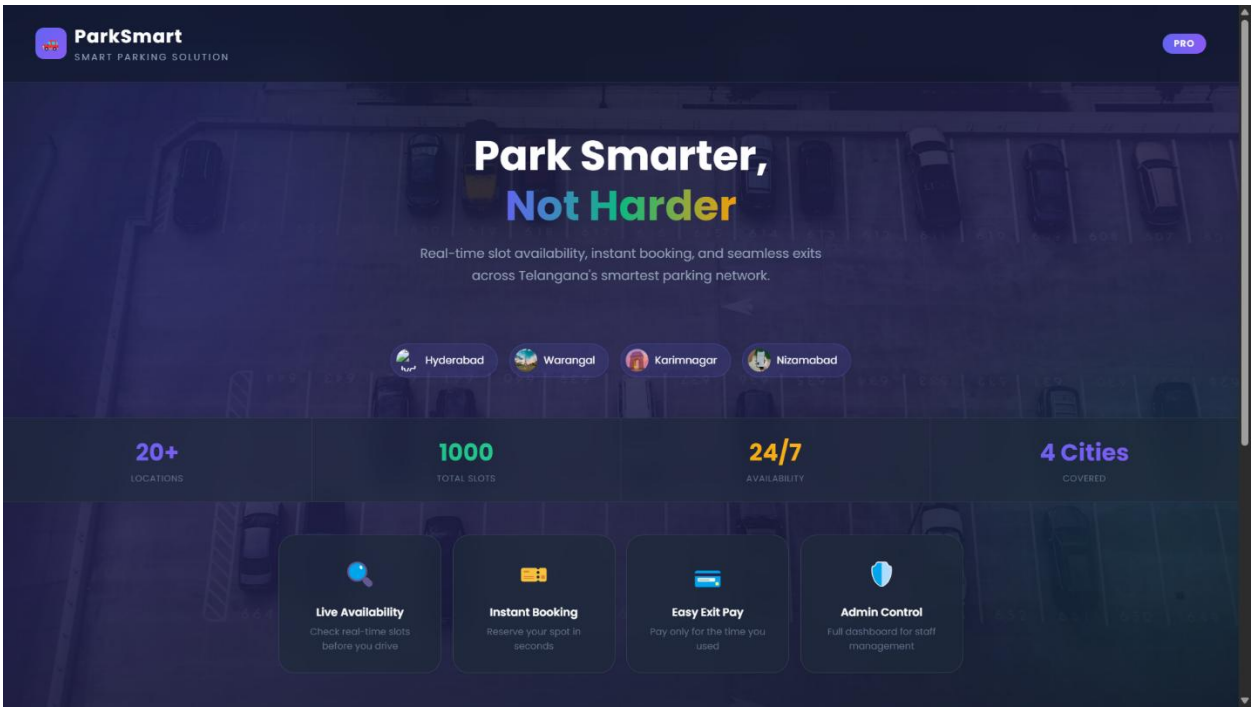
The payment system ensures that users only pay for the time they used an item. The administrative dashboard and all associated programs enhance the efficiency with which an administrator manages their lots through real-time reports. The JavaScript-based dynamic updates associated with this project enable the system to respond to user input and significantly minimize manual labor.

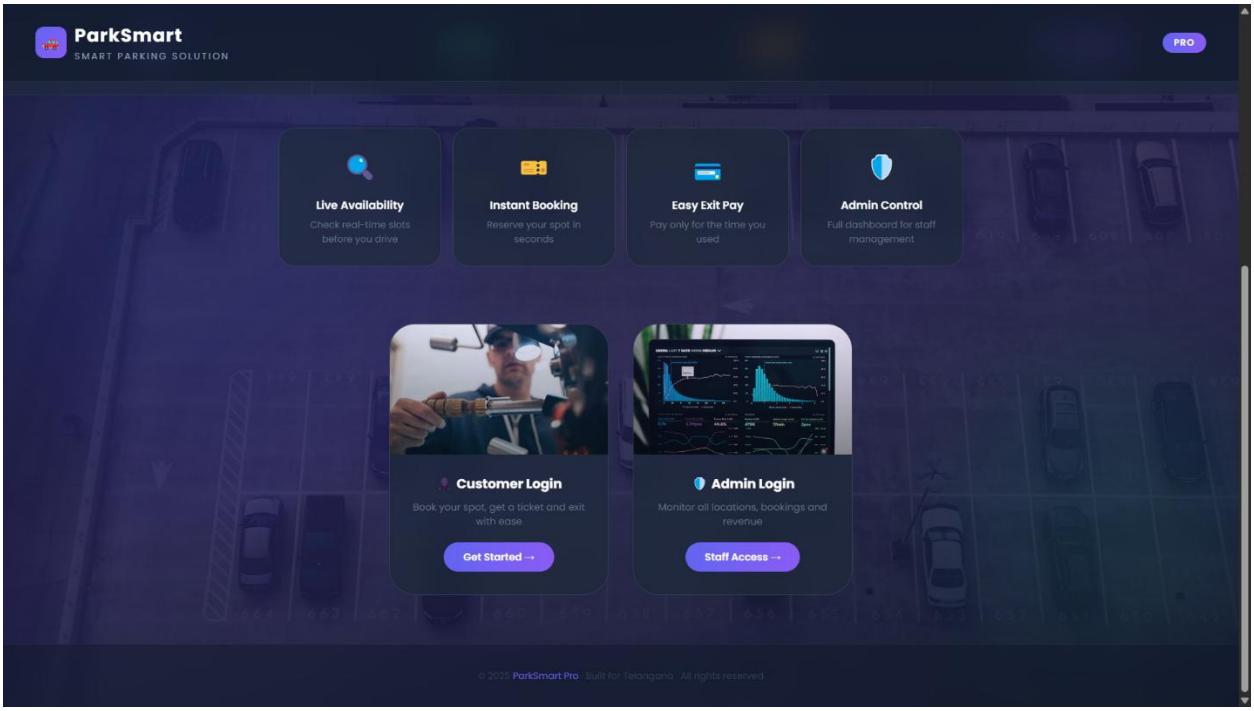
Even though using Local Storage for the purpose of demonstrating the project was sufficient, implementing the project with a cloud-based relational database would accommodate larger users and improve real-world performance.

RESULTS

Landing Page

Park Smart Pro's landing page serves as the primary user interface for accessing the smart parking system. The landing page contains the project's title ("Park Smarter, Not Harder"), and presents all major features of the system, including real-time availability of slots, instant booking, exit payment, and administrator control. The landing page will also allow users to select from among the four cities (Hyderabad, Warangal, Karimnagar, Nizamabad) as well as parking information (total number of locations, total number of slots, how many of those slots are available, and the four cities covered by the parking system). The landing page has been designed to promote easy navigation for both customers and administrators.

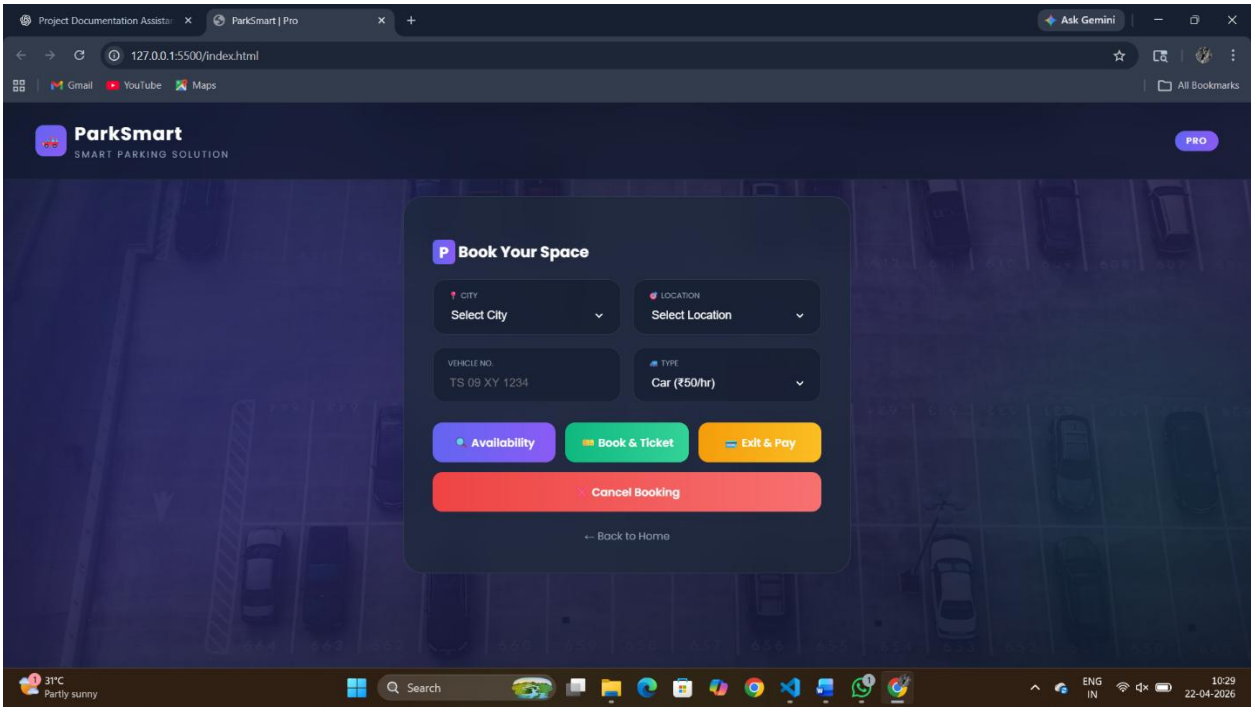




Customer Booking Page

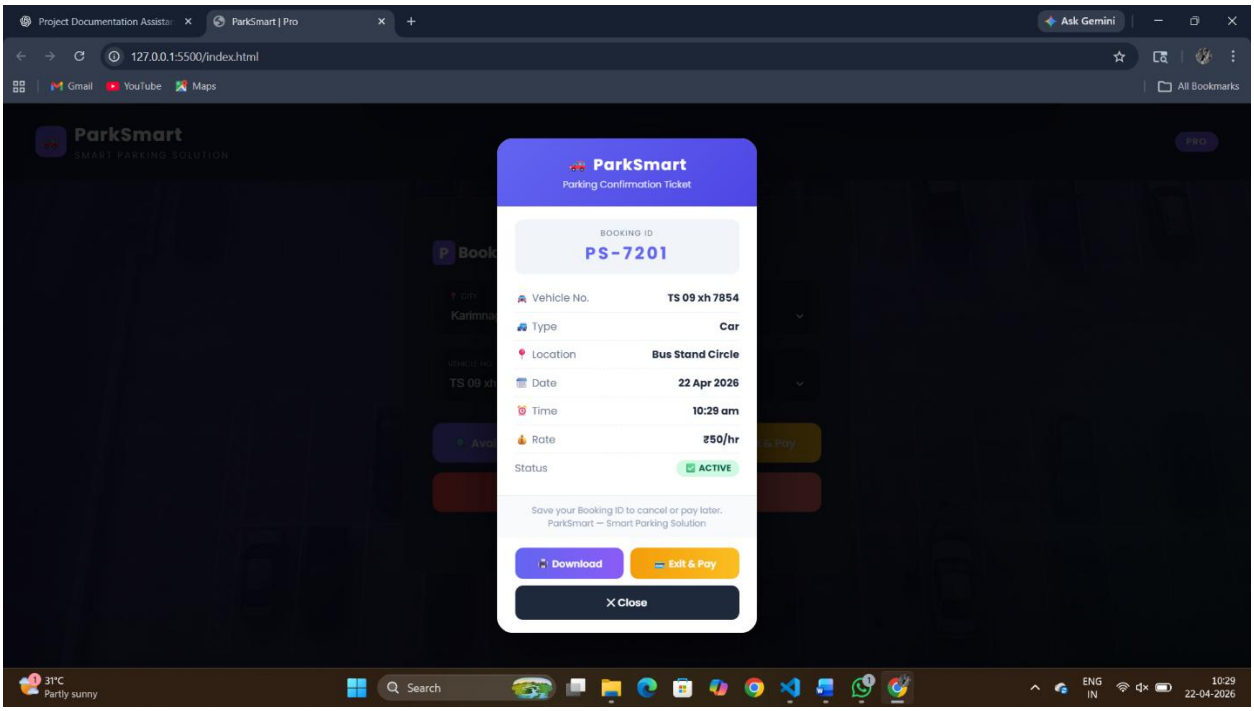
Customers can quickly and easily reserve a parking spot on the customer booking page. It allows customers to choose a city, choose the specific parking location, input their vehicle number and specify which type of car they have. The system gives users buttons to check for open slots, reserve a slot, make payments to exit the parking lot as well as cancel their reservations. As a result, the

customer booking page provides customers with an easy-to-use system for reserving a parking spot.



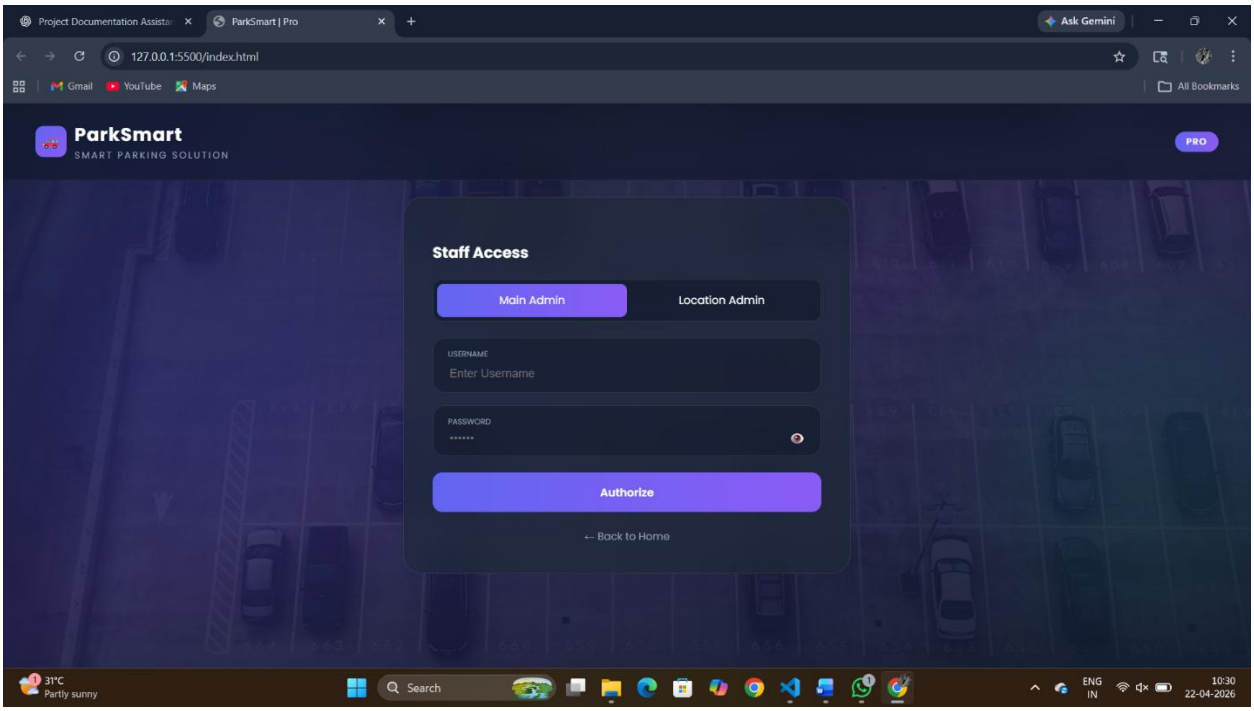
Digital Ticket Generation

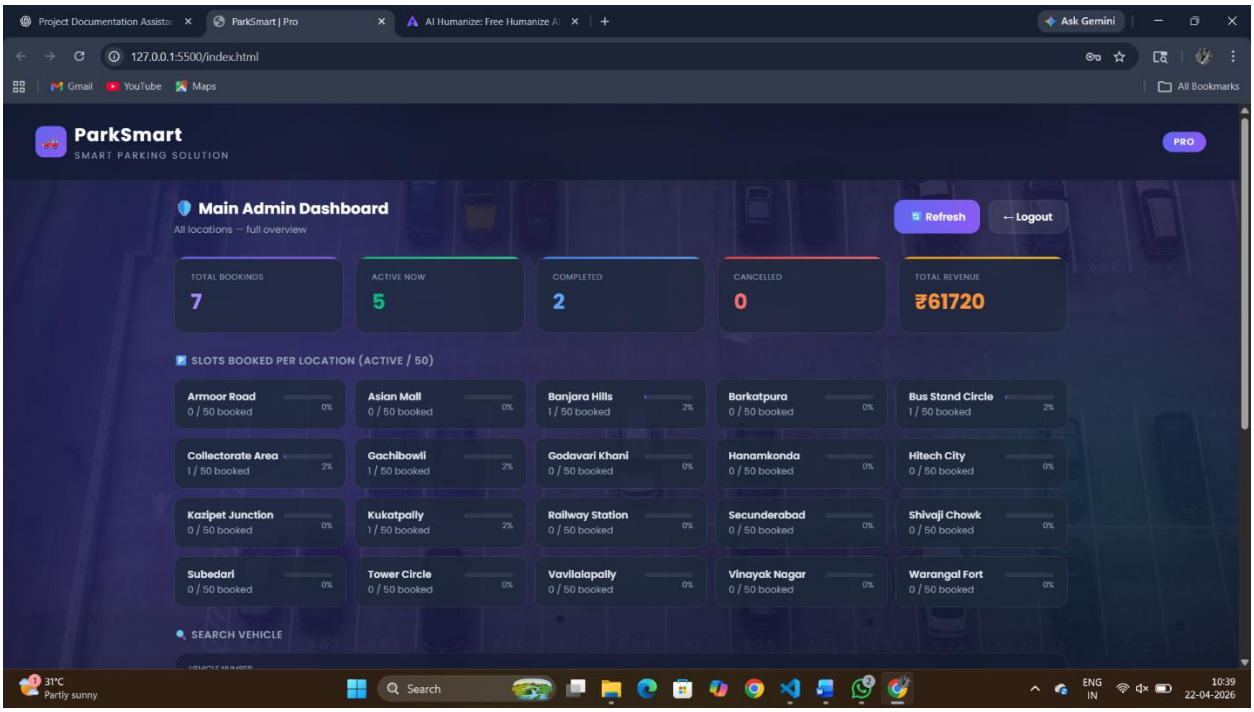
When you have successfully booked parking, you will receive a digital parking confirmation ticket created by the system. This ticket contains all the details of your booking, including Booking ID; Vehicle number; vehicle type; selected location; booking date; booking time; parking rate per hour; booking status. You must always carry this digital ticket as proof of your reservation and use it to assist you in completing your payment and exit.



Main Admin Login

A secure way to access the system administrator (who manages the entire system of all parking locations) is through the Main Admin Login page. The administrator uses their username and password to sign in to the parking management dashboard. The Main Admin has full control over bookings, parking slots, revenues, and the overall tracking of all parking locations in Telangana. This allows for better integration and management of parking locations throughout the state of Telangana.

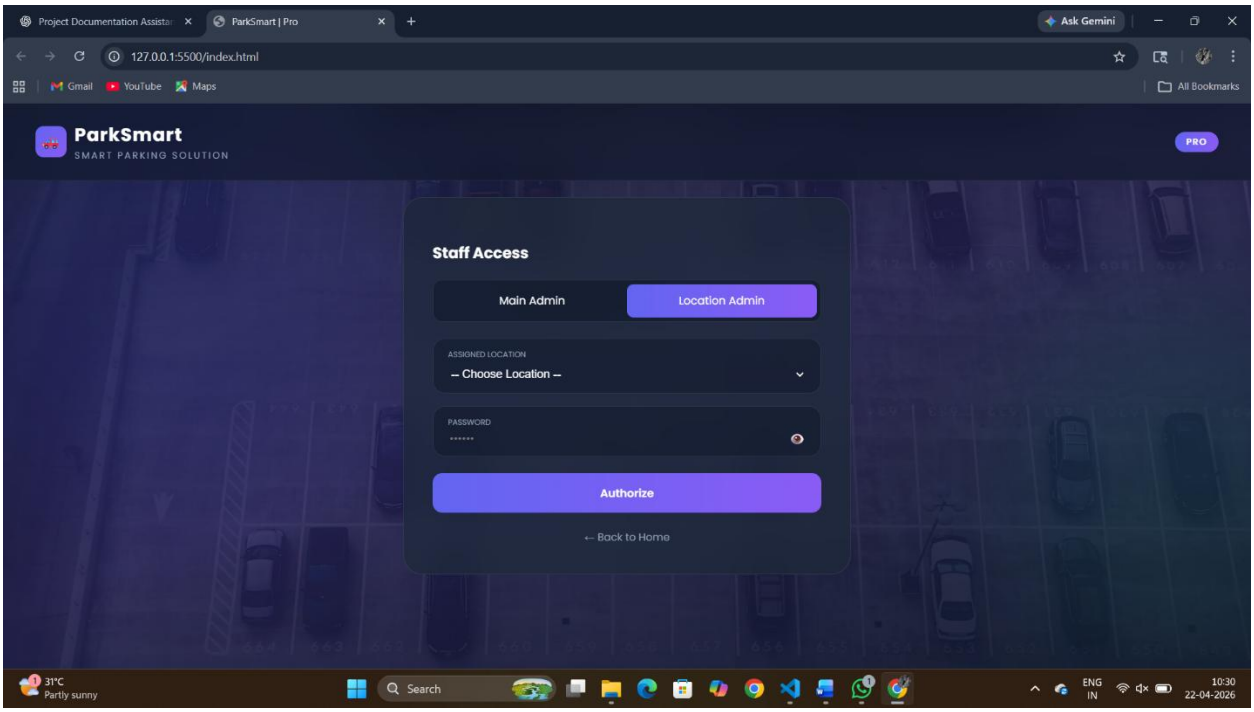




Location Admin Login

Staff who are assigned to a specific parking location will use this Location Admin login page. After the staff selects the assigned parking location, they will enter a password to log into to the location.

This role enables the staff to monitor local bookings, check the availability of parking spaces, and monitor revenues only for their parking area. Understanding how the various staff are controlling and managing the location-based parking lot helps provide effective parking management and staff control.



DISCUSSION

Park Smart Pro is a simple web-based solution to parking management problems. The project successfully digitized traditionally manual parking operations into a quick and convenient workflow through its use of HTML, CSS, JavaScript, and Local Storage.

The customer module provides ease of use while the admin module provides controls over parking operations. Additionally, the system assists in providing smart city solutions through its congestion reduction and mobility enhancement features.

Future enhancements could provide the project with additional functionality such as backend integration, GPS-driven parking recommendations, payment gateway support, and IoT-driven real-time space detection, making it an even better fit for the real world.

SCOPE OF THE PROJECT

The Parks Smart Pro is intended to give Intelligent Digital Parking Management Solution, to the Urban Areas of Telangana, will therefore reduce Traffic Congestion, Save Fuel, Improve Parking

Efficiency through real-time slot booking and Administrator Monitoring, and can be expanded to malls, Hospitals, Railway Stations, Airports, Smart City Projects, etc. in the future.

APPLICATIONS

Any area requiring parking management for multiple vehicles can utilize Park Smart Pro.

Some examples of such areas include shopping malls; airports; railway stations; hospitals; commercial buildings; educational institutions; apartment complexes; smart city-parking zones.

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